

JAVA LAB RECORD

Experiment No: 04

Date: 20/09/26

Title: Student Performance Analysis System Using Java

1. Objective / Aim

To design and implement a Java-based Student Performance Analysis System using object-oriented programming, arrays, constructors, methods, loops, conditional statements, and String operations to calculate total, average, highest and lowest marks, percentage, grade, result, and performance remarks.

2. System Requirements

Hardware Requirements	Software Requirements
Computer/Laptop	Operating System: Windows / Linux / macOS
Processor: Intel Core i3 or equivalent	JDK: 17 or 21
RAM: Minimum 4 GB	IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
Storage: Minimum 1 GB free space	Command Prompt / Terminal
Keyboard and Monitor	Java compiler (javac) and Java runtime (java)

3. Algorithm / Logic Description

1. Start the program.
2. Create a Scanner object to accept input from the user.
3. Read the student's roll number and name.
4. Create an integer array of size 5 to store marks for five subjects.
5. Read the five subject marks using a for loop.
6. Create a Student object by passing roll number, name, and marks to the constructor.
7. Calculate total marks by adding all elements of the marks array.
8. Calculate average by dividing total marks by the number of subjects.
9. Find the highest mark using Math.max() and the lowest mark using Math.min().
10. Calculate percentage using total marks and the maximum possible marks.
11. Determine grade using if-else conditions: A+, A, B, C, D, or F.
12. Determine PASS or FAIL using a 50% threshold.
13. Generate a performance remark based on the percentage.
14. Trim and convert the student name to uppercase and display its length.
15. Display the complete performance report and close the Scanner.
16. Stop the program.

4. UML Class Diagram

```
+-----+ | Student | +-----+ | - rollNumber :  
int | | - studentName : String | | - marks : int[] | +-----+ | + Student(int,  
String, int[]) | | + calculateTotal() : int | | + calculateAverage() : double | | + findHighest() : int | | +  
findLowest() : int | | + calculatePercentage() : double | | + determineGrade() : String | | + determineResult() :
```

```
String || + performanceRemark() : String || + displayDetails() : void |
+-----+ creates StudentPerformanceAnalysis -----> Student
+-----+ | + main(String[]) : void | +-----+
```

5. Source Code

```
import java.util.Scanner;

class Student {

    int rollNumber;
    String studentName;
    int[] marks;

    // Constructor
    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    // Calculate total marks
    int calculateTotal() {
        int total = 0;

        for (int mark : marks) {
            total += mark;
        }

        return total;
    }

    // Calculate average marks
    double calculateAverage() {
        return calculateTotal() / (double) marks.length;
    }

    // Find highest mark
    int findHighest() {
        int highest = marks[0];

        for (int mark : marks) {
            highest = Math.max(highest, mark);
        }

        return highest;
    }

    // Find lowest mark
    int findLowest() {
        int lowest = marks[0];

        for (int mark : marks) {
            lowest = Math.min(lowest, mark);
        }

        return lowest;
    }

    // Calculate percentage
    double calculatePercentage() {
        int total = calculateTotal();
        return (total / (marks.length * 100.0)) * 100;
    }

    // Determine grade
    String determineGrade() {
```

```

double percentage = calculatePercentage();

if (percentage >= 90)
    return "A+";
else if (percentage >= 80)
    return "A";
else if (percentage >= 70)
    return "B";
else if (percentage >= 60)
    return "C";
else if (percentage >= 50)
    return "D";
else
    return "F";
}

// Determine result
String determineResult() {
    if (calculatePercentage() >= 50)
        return "PASS";
    else
        return "FAIL";
}

// Performance remark
String performanceRemark() {
    double percentage = calculatePercentage();

    if (percentage >= 90)
        return "Excellent Performance";
    else if (percentage >= 80)
        return "Very Good Performance";
    else if (percentage >= 70)
        return "Good Performance";
    else if (percentage >= 60)
        return "Average Performance";
    else if (percentage >= 50)
        return "Needs Improvement";
    else
        return "Poor Performance";
}

// Display complete student details
void displayDetails() {

    String formattedName = studentName.trim().toUpperCase();

    double percentage = calculatePercentage();

    // Round percentage to two decimal places
    double roundedPercentage = Math.round((percentage * 100.0) / 100.0);

    System.out.println("\n===== STUDENT PERFORMANCE REPORT =====");
    System.out.println("Roll Number      : " + rollNumber);
    System.out.println("Student Name      : " + formattedName);
    System.out.println("Name Length       : " + formattedName.length());

    System.out.println("Marks:");
    for (int i = 0; i < marks.length; i++) {
        System.out.println("Subject " + (i + 1) + "      : " + marks[i]);
    }

    System.out.println("Total Marks      : " + calculateTotal());
    System.out.println("Average Marks     : " + calculateAverage());
    System.out.println("Highest Marks    : " + findHighest());
    System.out.println("Lowest Marks     : " + findLowest());
    System.out.println("Percentage       : " + roundedPercentage + "%");
}

```

```

        System.out.println("Grade           : " + determineGrade());
        System.out.println("Result          : " + determineResult());
        System.out.println("Performance      : " + performanceRemark());

        System.out.println("=====");
    }
}

public class StudentPerformanceAnalysis {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();

        sc.nextLine(); // Consume newline

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        int[] marks = new int[5];

        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < marks.length; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        // Create Student object
        Student student = new Student(rollNumber, studentName, marks);

        // Display performance report
        student.displayDetails();

        sc.close();
    }
}

```

6. Completion and Execution Commands

1. Save the program as: StudentPerformanceAnalysis.java 2. Open Command Prompt / Terminal and move to the folder containing the file. 3. Compile: javac StudentPerformanceAnalysis.java 4. Execute: java StudentPerformanceAnalysis Important: The public class name is StudentPerformanceAnalysis, so the file name must be StudentPerformanceAnalysis.java.

7. Output

```

Enter Roll Number: 101 Enter Student Name: Rahul Kumar Enter marks for 5 subjects: Subject 1: 95
Subject 2: 88 Subject 3: 92 Subject 4: 76 Subject 5: 89 ===== STUDENT PERFORMANCE REPORT
===== Roll Number : 101 Student Name : RAHUL KUMAR Name Length : 11 Marks: Subject 1 : 95 Subject
2 : 88 Subject 3 : 92 Subject 4 : 76 Subject 5 : 89 Total Marks : 440 Average Marks : 88.0 Highest Marks :
95 Lowest Marks : 76 Percentage : 88.0% Grade : A Result : PASS Performance : Very Good Performance
=====

```

Result

The Student Performance Analysis System was successfully designed, compiled, and executed. The program accepts student details and five subject marks and generates a complete performance report.